

Python_Test

File Edit Selection View Go Run Terminal Help

EXPLORER

PYTHON_TEST

contrast_enhancement_comparison.png

create_challenge_images.py

create_test_image_day3.py

create_test_image.py

crop.py

custom_cyberpunk.png

custom_mask.png

custom_sketch.png

custom_vintage.png

day5_transformation.py

day6_arithmetic_bitwise.py

day7_mini_challenge.py

day8_colour_spaces.py

day9_histogram.py

day10_smoothing_blurring.py

day11_thresholding.py

dissolve.mp4

drawin_basics.py

drawing.py

edge_preserving_comparison.png

equalization_comparison.png

equalize.py

equalized_image.png

ex_1_day_3.py

ex_2_day_3.py

ex_3_day_3.py

ex_4_day_3.py

exercise_1_custom_pipeline.py

exercise_1_house.py

exercise_1.py

exercise_2_comparison.py

OUTLINE

TIMELINE

kernel.py

exercise4_edge_preserving.py

exercise5_deblurring.py

blurring.py

day11_thresholding.py

CHAT

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

def interactive_threshold(image_gray):

Create trackbars

methods = ["Binary", "Binary Inv", "Trunc", "Tozero"]

print("\nInteractive Threshold Explorer:")

print(" - Use trackbars to adjust threshold and method")

print(" - Press ESC to exit")

while True:

T = cv2.getTrackbarPos("Threshold", "Threshold Explorer")

method_idx = cv2.getTrackbarPos("Method", "Threshold Explorer")

method_map = {

0: cv2.THRESH_BINARY,

1: cv2.THRESH_BINARY_INV,

2: cv2.THRESH_TRUNC,

3: cv2.THRESH_TOZERO

}

_, result = cv2.threshold(image_gray, T, 255, method_map[method_idx])

Convert to color for display

result_color = cv2.cvtColor(result, cv2.COLOR_GRAY2BGR)

original_color = cv2.cvtColor(image_gray, cv2.COLOR_GRAY2BGR)

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

- No manual tuning needed

- Optimal for bimodal histograms (two distinct peaks)

Disadvantages:

- Fails if histogram is not bimodal

- Assumes uniform lighting

[Section 5] All Thresholding Types Comparison

[Section 6] Practical Application: Coin Segmentation

Found 5 coins!

Do you want to install the recommended 'Python' extension from Microsoft for the Python language?

Install

Show Recommendations

Ln 252, Col 1

Spaces: 4

UTF-8

CRLF

{ }

Python

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16:00

16-06-2026

Python_Test

EXPLORER

- PYTHON_TEST
 - ex_4_day_3.py
 - exercise1_custom_pipeline.py
 - exercise1_house.py
 - exercise1.py
 - exercise2_comparison.py
 - exercise2_target.py
 - exercise2.py
 - exercise3_chessboard.py
 - exercise3.py
 - exercise4_annotation.py
 - exercise5_clock.py
 - exercise1_brightness_contrast.py
 - exercise1_channel_art.py
 - exercise1_histogram_analyzer.py
 - exercise1_interactive_blur.py
 - exercise1_threshold_optimizer.py**
 - exercise1_thumbnail_gallery.py
 - exercise2_color_picker.py
 - exercise2_equalization_grid.py
 - exercise2_image_dissolve.py
 - exercise2_noise_reduction.py
 - exercise2_panorama_prep.py
 - exercise3_3d_color_histogram.py
 - exercise3_color_isolation.py
 - exercise3_custom_kernel.py
 - exercise3_mask_tool.py
 - exercise3_rotation_animation.py
 - exercise4_color_space_visualiser.py
 - exercise4_colour_splash.py
 - exercise4_edge_preserving.py
 - exercise4_histogram_search.py

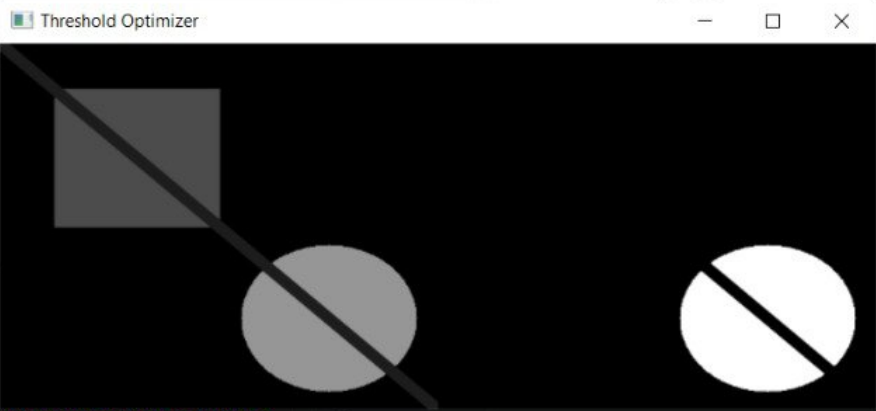
exercise1_threshold_optimizer.py

```
4 def threshold_optimizer(image_path):
41
42     # Apply threshold
43     _, thresholded = cv2.threshold(gray, optimal_T, 255, cv2.THRESH_BINARY)
44
45     # Show results
46     comparison = np.hstack([
47         cv2.resize(gray, (300, 250)),
48         cv2.resize(thresholded, (300, 250))
49     ])
50
51     cv2.putText(comparison, f"Optimal T = {optimal_T}", (320, 30), cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0))
52
53     cv2.imshow("Threshold Optimizer", comparison)
54     cv2.waitKey(0)
55
56     return optimal_T
57
58 # Run optimizer
59 threshold_optimizer("test_image.png")
60 cv2.destroyAllWindows()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

hape_base.py", line 365, in hstack
return _nx.concatenate(arrs, 1, dtype=dtype, casting=casting)
~~~~~  
ValueError: all the input arrays must have same number of dimensions, but the array at index 0 has 2 dimension(s) and the array at index 2 has 3 dimension(s)  
PS C:\Users\jebbron\jason\Desktop\Python\_Test> python exercise1\_threshold\_optimizer.py  
Detected two peaks at 76 and 150  
Optimal threshold T = 77  
PS C:\Users\jebbron\jason\Desktop\Python\_Test> python exercise1\_threshold\_optimizer.py  
Detected two peaks at 76 and 150  
Optimal threshold T = 77

Threshold Optimizer



powershell powershell powershell powershell powershell powershell python

Describe what to build

Local Default Approvals

Ln 56, Col 5 Spaces: 4 UTF-8 CRLF Python

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Python\_Test

Sign In

EXPLORER

PYTHON\_TEST

exercise\_1.py

exercise\_2\_comparison.py

exercise\_2\_target.py

exercise\_2.py

exercise\_3\_chessboard.py

exercise\_3.py

exercise\_4\_annotation.py

exercise\_5\_clock.py

exercise1\_brightness\_contrast.py

exercise1\_channel\_art.py

exercise1\_histogram\_analyzer.py

exercise1\_interactive\_blur.py

exercise1\_threshold\_optimizer.py

exercise1\_thumbnail\_gallery.py

exercise2\_color\_picker.py

exercise2\_document\_binarization.py

exercise2\_equalization\_grid.py

exercise2\_image\_dissolve.py

exercise2\_noise\_reduction.py

exercise2\_panorama\_prep.py

exercise3\_3d\_color\_histogram.py

exercise3\_color\_isolation.py

exercise3\_custom\_kernal.py

exercise3\_mask\_tool.py

exercise3\_rotation\_animation.py

exercise4\_color\_space\_visualiser.py

exercise4\_colour\_splash.py

exercise4\_edge\_preserving.py

exercise4\_histogram\_search.py

exercise4\_smart\_crop.py

exercise5\_contrast\_enhancement.py

OUTLINE

TIMELINE

exercise2\_document\_binarization.py

```
103 for name, doc in docs:
106     comparison = np.hstack([
109         ]
110
111     # Since comparison is grayscale, use a grayscale color value
112     cv2.putText(
113         comparison,
114         name,
115         (10, 30),
116         cv2.FONT_HERSHEY_SIMPLEX,
117         0.7,
118         255,
119         2
120     )
121
122     cv2.imshow(f"Document Binarization - {name}", comparison)
123     cv2.waitKey(0)
124
125 cv2.destroyAllWindows()
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
\jebon jason\Desktop\Python_Test\exercise2_document_binarization.py
PS C:\Users\jebon jason\Desktop\Python_Test> python exercise2_document_binarization.py
C:\Users\jebon jason\AppData\Local\Programs\Python\Python314\python.exe: can't open file 'C:\Users\jebon jason\Desktop\Python_Test\exercise2_document_binarization.py': [Errno 2] No such file or directory
PS C:\Users\jebon jason\Desktop\Python_Test> python exercise_2_document_binarization.py
C:\Users\jebon jason\AppData\Local\Programs\Python\Python314\python.exe: can't open file 'C:\Users\jebon jason\Desktop\Python_Test\exercise_2_document_binarization.py': [Errno 2] No such file or directory
PS C:\Users\jebon jason\Desktop\Python_Test> python exercise2_document_binarization.py
```

Document Binarization - Normal

Normal

Normal Document Text

Normal Document Text

Build with Agent

AI responses may be inaccurate

Generate Agent Instructions to onboard AI onto your codebase.

powershell

python

Describe what to build

Local Default Approvals

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File Edit Selection View Go Run Terminal Help Python\_Test

EXPLORER

- PYTHON\_TEST
  - exercise1\_channel\_art.py
  - exercise1\_histogram\_analyzer.py
  - exercise1\_interactive\_blur.py
  - exercise1\_threshold\_optimizer.py
  - exercise1\_thumbnail\_gallery.py
  - exercise2\_color\_picker.py
  - exercise2\_document\_binarization.py
  - exercise2\_equalization\_grid.py
  - exercise2\_image\_dissolve.py
  - exercise2\_noise\_reduction.py

exercise1\_threshold\_optimizer.py exercise2\_document\_binarization.py exercise3\_multi\_object\_segmentation.py

```
111
112 test_objects = cv2.add(test_objects, noise)
113
114 cv2.imshow("Original Objects", test_objects)
115 cv2.waitKey(0)
116
117 # Segment objects
118 result, count, binary = segment_objects(
119     test_objects,
120     min_area=200
121 )
```

Segmented Objects (Binary Mask)

Original Objects

Terminal

```
PS C:\Users\jeb_\Desktop\Python_Test> python exercise2_document_binarization.py
PS C:\Users\jeb_\Desktop\Python_Test> python exercise3_multi_object_segmentation.py
Objects detected: 6
```

flipping.py getting\_and\_setting.py

OUTLINE

TIMELINE

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Python\_Test

File Edit Selection View Go Run Terminal Help

EXPLORER

- exercise2\_document\_binarization.py
- exercise3\_multi\_object\_segmentation.py
- exercise4\_background\_removal.py

Mask

Background Removed

Composite (Red Background)

np.where(  
: np.newaxis] == 255,  
bg  
Composite (Red Background)", composite)  
0)  
"background\_removed.png", result)  
: background\_removed.png")  
: Could not load test\_image.png")  
ndows()

exercise4\_  
exercise4\_colour\_splash.py  
exercise4\_edge\_preserving.py  
exercise4\_histogram\_search.py  
exercise4\_smart\_crop.py  
exercise5\_contrast\_enhancement.py  
exercise5\_deblurring.py  
exercise5\_heart\_mask.py  
exercise5\_image\_alignment.py  
Figure\_1.png  
final\_collage.png  
first\_image.py  
flipping.py  
getting\_and\_setting.py  
gradient.png

OUTLINE

TIMELINE

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

C:\Users\jebroon jason\AppData\Local\Programs\Python\Python314\python.exe: can't open file 'C:\\Users\\jebroon jason\\Desktop\\Python\_Test\\exercise\_2\_document\_binarization.py': [Errno 2] No such file or directory  
PS C:\Users\jebroon jason\Desktop\Python\_Test> python exercise\_2\_document\_binarization.py  
C:\Users\jebroon jason\AppData\Local\Programs\Python\Python314\python.exe: can't open file 'C:\\Users\\jebroon jason\\Desktop\\Python\_Test\\exercise\_2\_document\_binarization.py': [Errno 2] No such file or directory  
PS C:\Users\jebroon jason\Desktop\Python\_Test> python exercise2\_document\_binarization.py  
PS C:\Users\jebroon jason\Desktop\Python\_Test> python exercise3\_multi\_object\_segmentation.py  
Objects detected: 6  
PS C:\Users\jebroon jason\Desktop\Python\_Test> python exercise4\_background\_removal.py

Original

Describe what to build

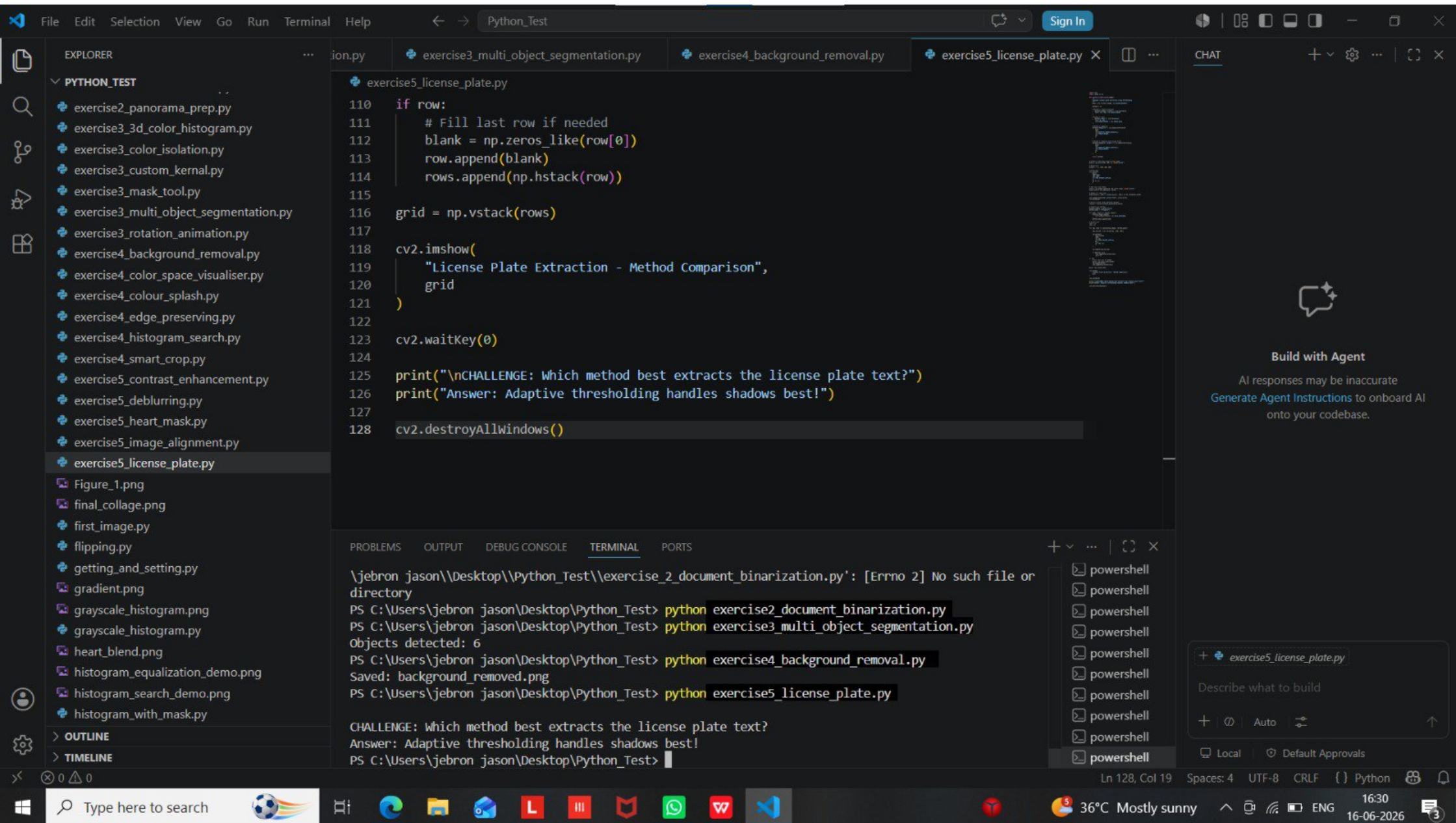
+ | | Auto | |

Local | Default Approvals

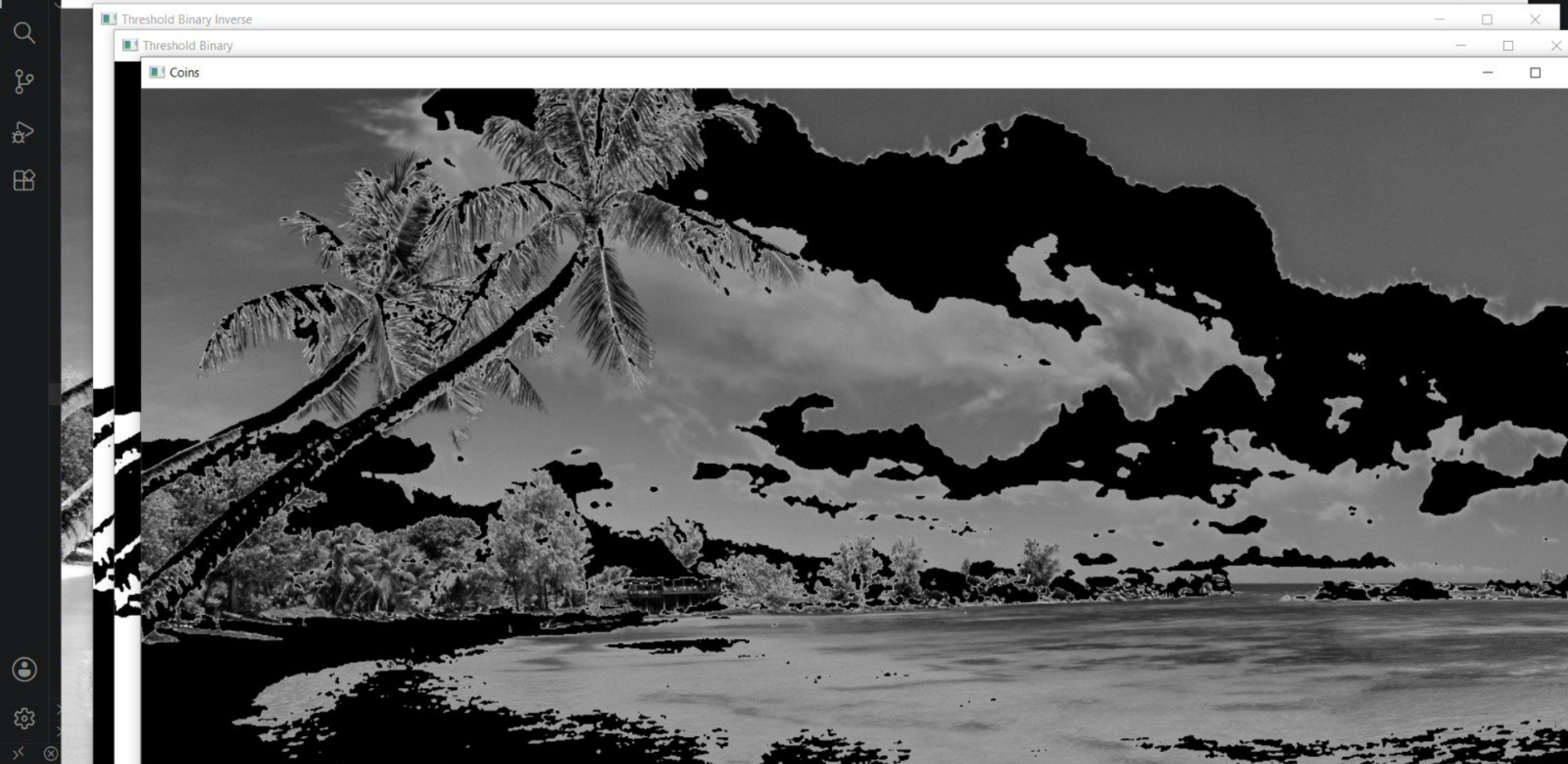
Ln 89, Col 24 Spaces: 4 UTF-8 CRLF {} Python

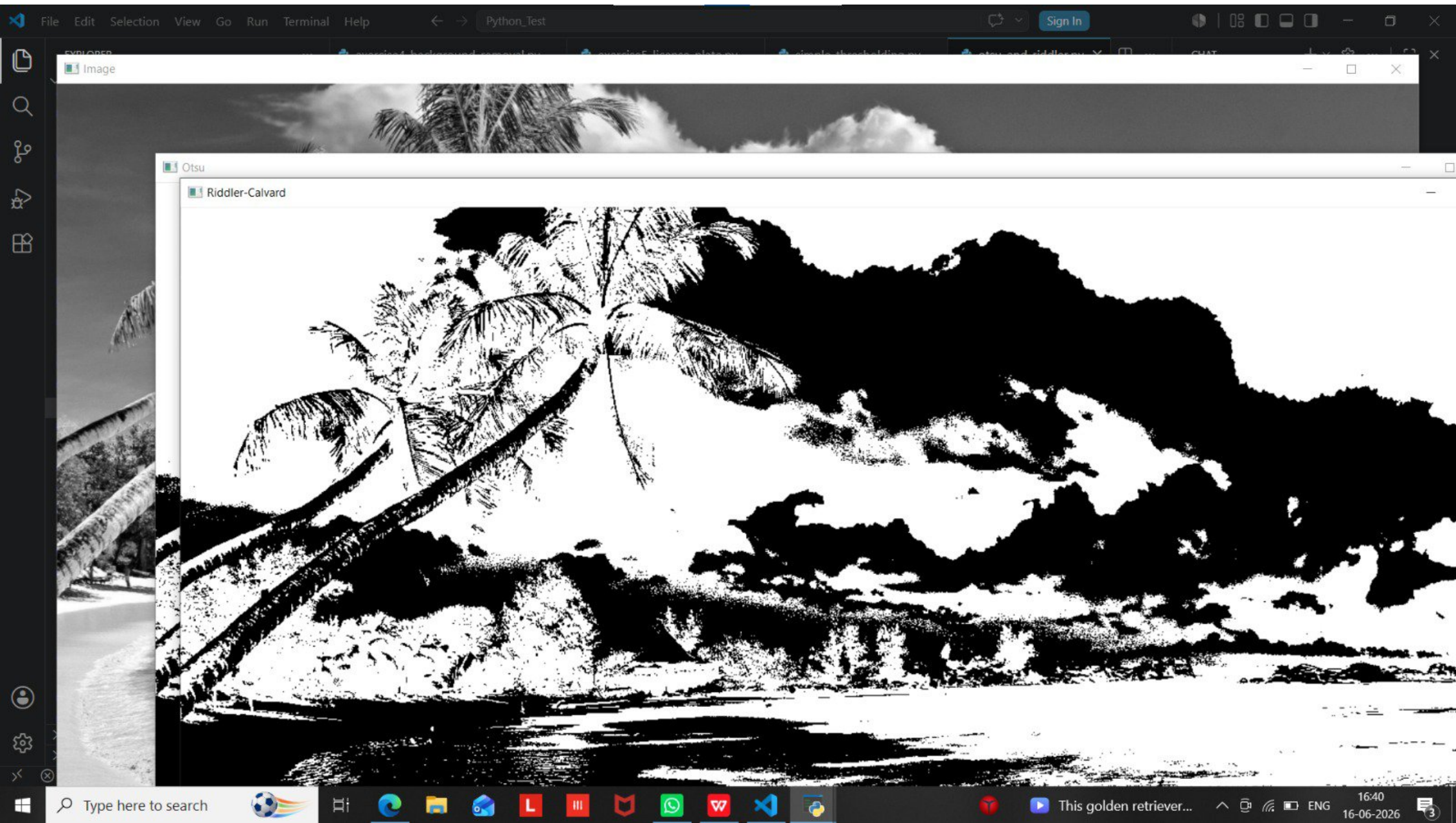
Type here to search

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Mean Thresh

Gaussian Thresh

